

Regulation of brown adipose tissue development and white fat reduction by L-arginine.

PURPOSE OF REVIEW: Brown adipose tissue (BAT), which is present in humans, plays an important role in oxidation of fatty acids and glucose. The purpose of this review is to highlight an important role for L-arginine in regulating BAT growth and development, thereby reducing obesity in mammals. RECENT FINDINGS: Dietary supplementation with L-arginine reduces white adipose tissue in genetically or diet-induced obese rats, obese pregnant sheep, and obese humans with type II diabetes. L-arginine treatment enhances BAT growth in both fetuses and postnatal animals. At molecular and cellular levels, L-arginine stimulates expression of peroxisome proliferator-activated receptor- γ coactivator 1 (the master regulator of mitochondrial biogenesis), nitric oxide synthase, heme oxygenase, and adenosine monophosphate-activated protein kinase. At the whole body level, L-arginine increases blood flow to insulin-sensitive tissues, adipose tissue lipolysis, and the catabolism of glucose and fatty acids, but inhibits fatty acid synthesis and ameliorates oxidative stress, thereby improving metabolic profile. SUMMARY: L-arginine increases mammalian BAT growth and development via mechanisms involving gene expression, nitric oxide signaling, and protein synthesis. This enhances the oxidation of energy substrates and, thus, reduces white fat accretion in the body. L-arginine holds great promise in preventing and treating obesity in humans.